Assignment 14

TypeScript OOP Assignment – Class Diagram

Part A - System: Simple Notes Management System (7 Marks)

1. Base Class – User (1 Mark)

Create a class called User.
The class should contain the following properties:

- Id (number)
- **name** (String)
- **email** (String)
- **Password** (String)
- **Phone** (String)
- **age** (Number) (Must be between 18 and 60)

Requirements:

- Use appropriate access modifiers (public, private, protected).
- Create a method called displayInfo() that prints the user's information.

2. Inheritance – Admin User (1 Mark)

Create a class called Admin that extends the User class.

Requirements:

- The Admin class should include a method that allows the admin to manage notes.
- Apply the concept of **inheritance**.

3. Class - Note (1 Mark)

Create a class called Note

- **Id** (number)
- **tittle**(String)
- **content**(String)
- userId(reference to User)

Requirements:

- Create a method called preview() that returns a short preview of the note content.

4. Composition– NoteBook and Notes (1 Mark)

Create a class called NoteBook

Requirements:

- The NoteBook class should contain a collection of Notes objects .
- Implement methods such as : addNote() , removeNote()
- The relationship between Notebook and Note must represent Composition .

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5. Aggregation – User and Notebook (1 Mark)

Create a relationship between:

- User
- NoteBook

Requirements:

- A user can own multiple notebooks.
- This relationship should represent **Aggregation**.

6. Association – Note and User (1 Mark)

Create a relationship between:

- User
- Note

Requirements:

- Each Note must have an author (the user who created it).
- This relationship should represent **Association**.

7. Generics– Data Storage(1 Mark)

Create a Generic Class called :

- Storage

Requirements:

- The class should be able to store any type of data .

Example operations may include :

1. **addItem ()**
2. **removeItem() .**
3. **getAllItems() .**

Part B - UML Digram (3 Markes)

1. Draw the Class Diagram (2 Mark)

Draw a UML Class Diagram that includes :

- **All classes**
- **Attributes**
- **Methods**
- **Relationships between classes**

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2. Identify Relationships (1 Mark)

Clearly identify the following relationships in your diagram:

- **Inheritance .**
- **Composition** .
- **Aggregation** .
- **Association** .